Report 4.11 – Risultati per k con $TIMES = 10$

2 maggio 2026

1 Grafici dei risultati per k e $TIMES = 10$

I grafici mostrano rispettivamente l'andamento di:

- Avg fQ
- fQ Min Found
- Profit Found
- Weight Found
- Profit GAP
- Weight GAP
- n of fQ Min Found

al variare di $k \in \{1000, 2000, 3000, 4000, 5000\}$, confrontando i risultati usando QALS e NON utilizzando QALS.

I risultati riportati sono stati ottenuti utilizzando:

- $\lambda = \frac{\sum_{i=1}^{n} p_i}{3}$
- l'istanza `ksp_3.txt`
- `TIMES = 10`

Profit ottimo	2679
Weight ottimo	496
TIMES	10

Tabella 1: Soluzione di riferimento del problema di knapsack.

2 Tabella riassuntiva

k	QALS							NO QALS						
	Avg fQ	fQ Min	Profit	Weight	Profit GAP	Weight GAP	n fQ Min	Avg fQ	fQ Min	Profit	Weight	Profit GAP	Weight GAP	n fQ Min
1000	17350642347.4	8241260668.67	3423	2275	-744	-1779	1	-714267512.7	-714267662	2483	501	196	-5	6
2000	18615897688.77	12244772130.67	3481	2635	-802	-2139	1	-714267601.6	-714267662	2483	501	196	-5	8
3000	17038731132.6	12039123606.67	4212	2618	-1533	-2122	1	-714267631.8	-714267662	2483	501	196	-5	9
4000	20763157976.07	16020029113.67	4249	2926	-1570	-2430	1	-714267662	-714267662	2483	501	196	-5	10
5000	17864058550.1	6034515014.67	2873	2041	-194	-1545	1	-714267662	-714267662	2483	501	196	-5	10

Tabella 2: Risultati numerici per $k \in \{1000, 2000, 3000, 4000, 5000\}$ con `TIMES = 10`.

3 Grafici

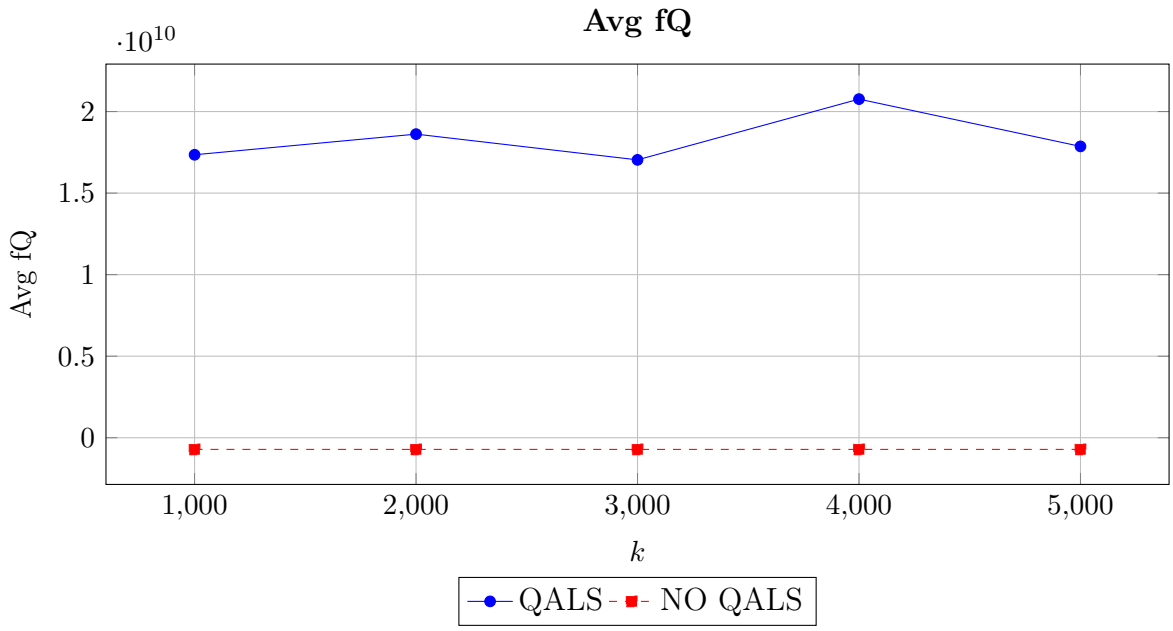

Figura 1: Andamento di Avg fQ al variare di k.

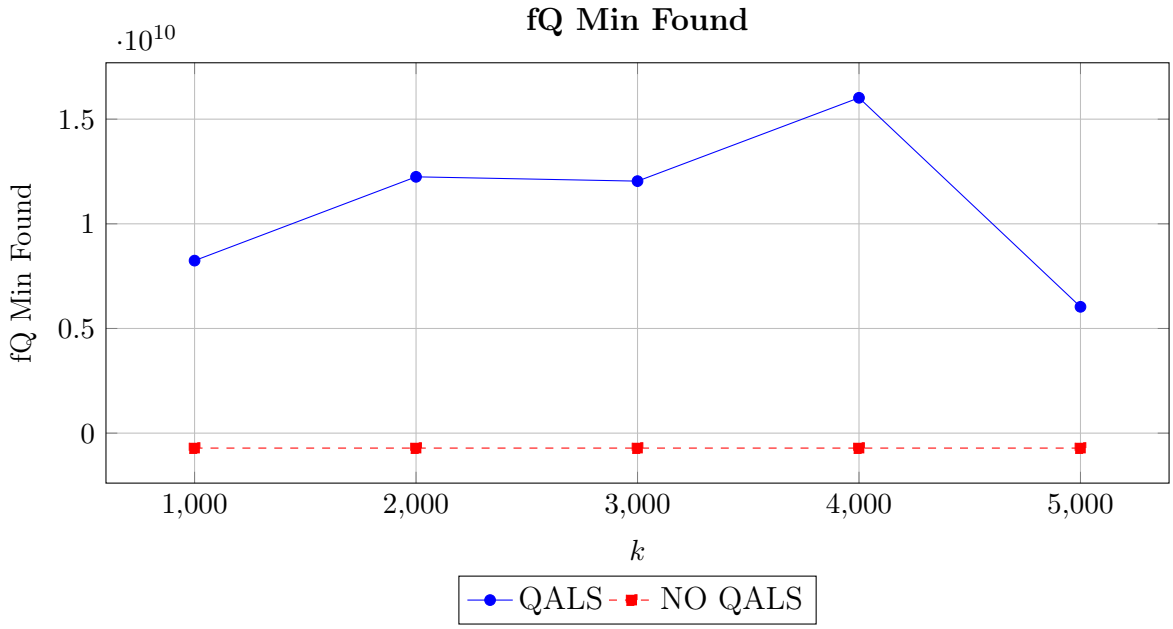

Figura 2: Andamento di fQ Min Found al variare di k.

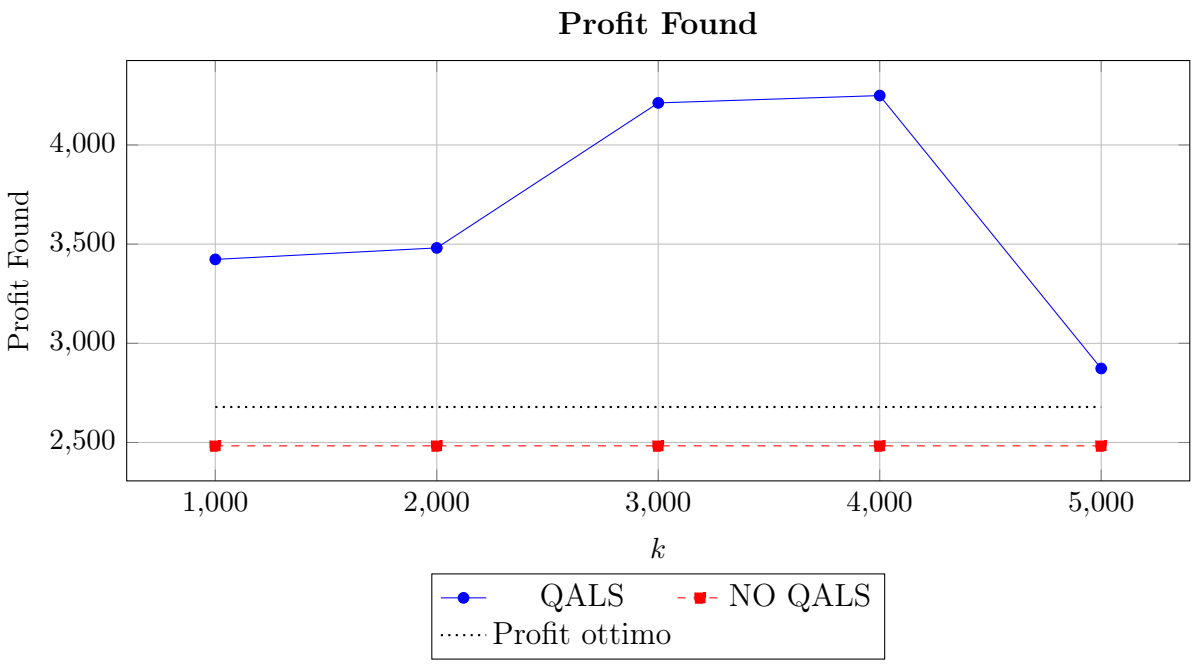

Figura 3: Andamento del profitto trovato al variare di k.

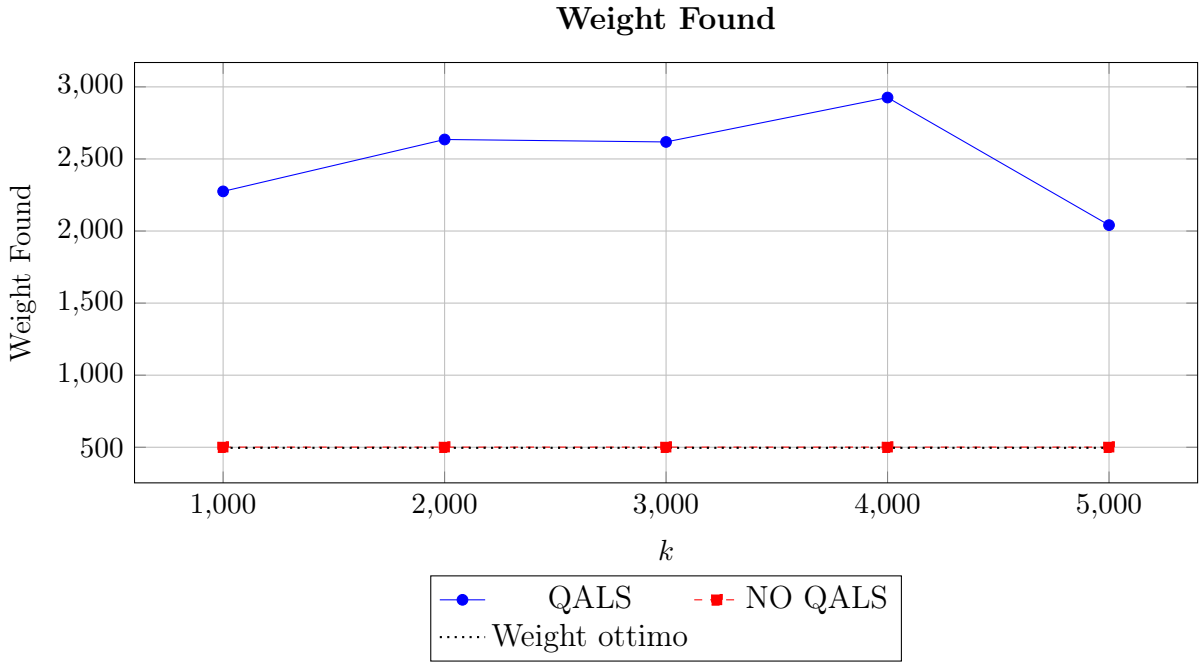

Figura 4: Andamento del peso trovato al variare di k.

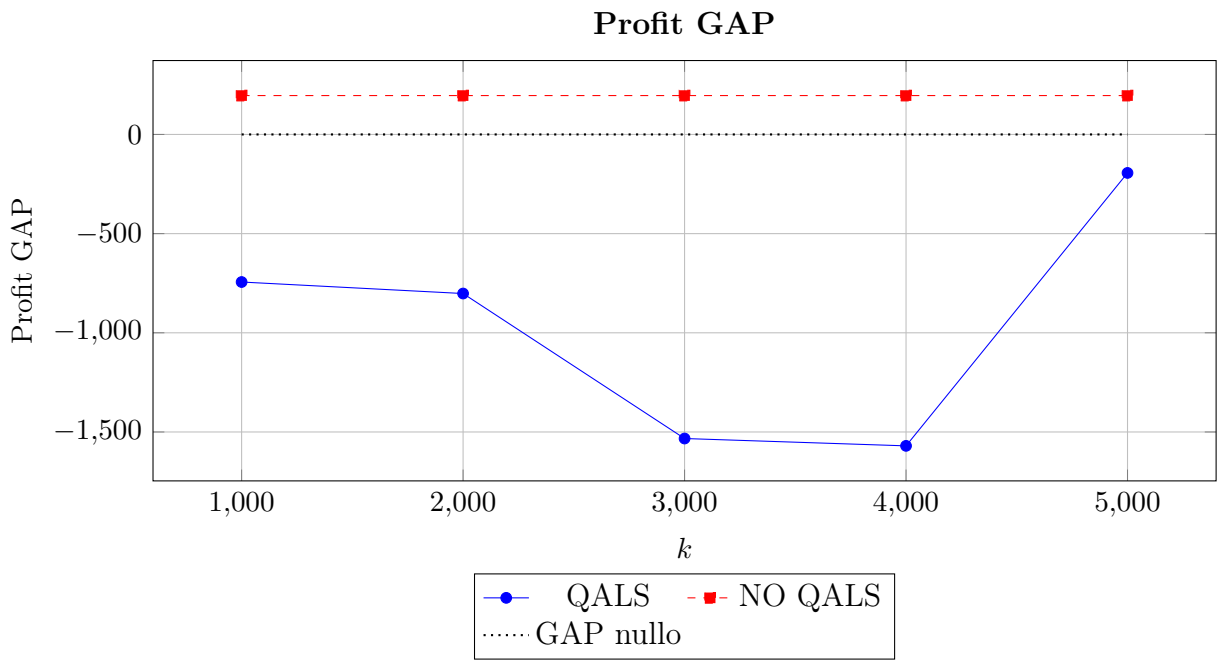

Figura 5: Andamento del Profit GAP al variare di k.

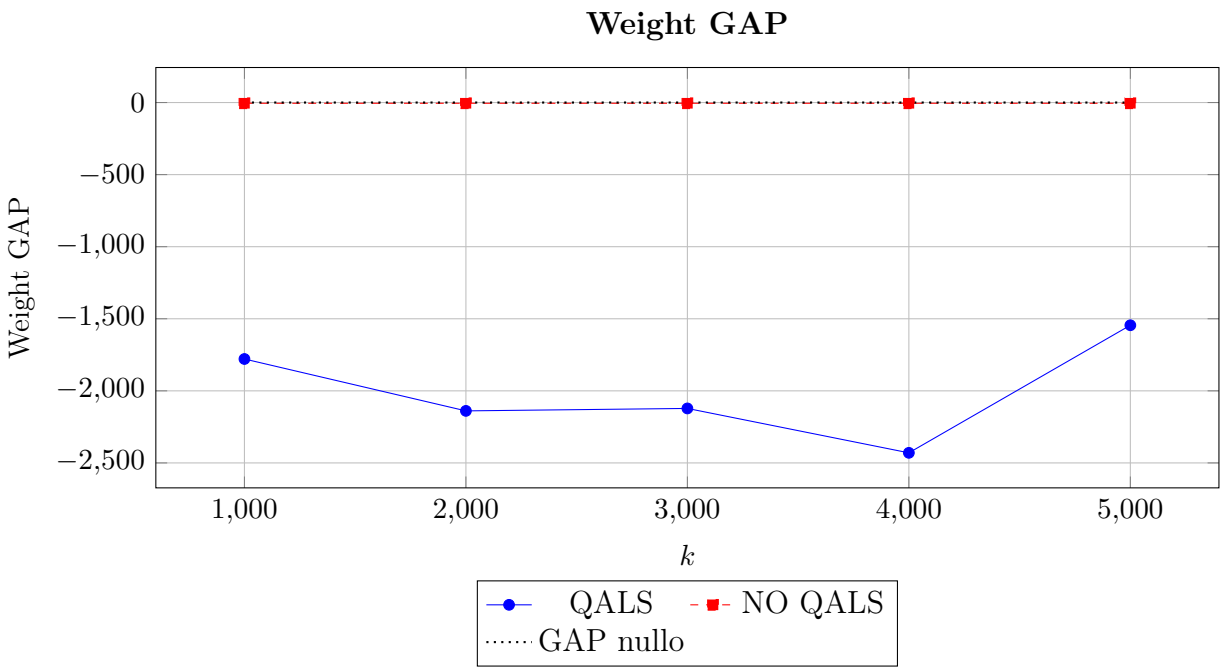

Figura 6: Andamento del Weight GAP al variare di k.

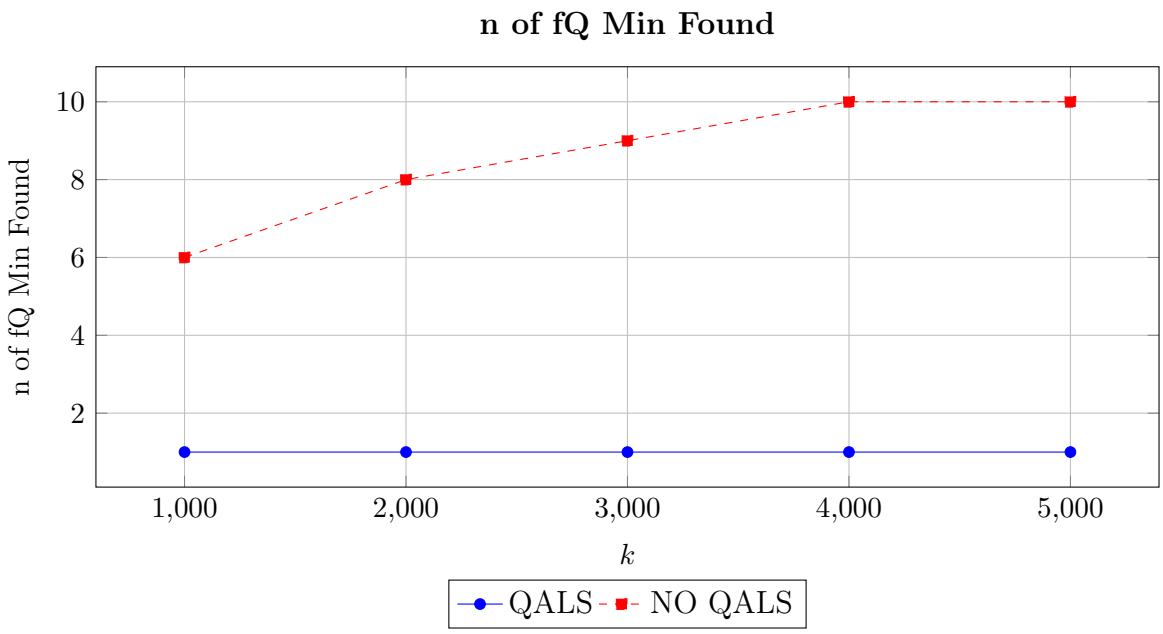

Figura 7: Numero di volte in cui è stato trovato il minimo fQ al variare di k.

4 Soluzioni trovate

In questa sezione sono riportati anche i vettori degli item selezionati per ogni valore di k.

k	Metodo	Items
1000	QALS	[1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1]
1000	NO QALS	[0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 1]
2000	QALS	[1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1, 0, 1, 0, 0, 1, 1, 1]
2000	NO QALS	[0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 1]
3000	QALS	[0, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1]
3000	NO QALS	[0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 1]
4000	QALS	[0, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1]
4000	NO QALS	[0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 1]
5000	QALS	[0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0]
5000	NO QALS	[0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 1]

Tabella 3: Vettori degli item trovati da QALS e NO QALS.